

KAIHUAI DENG

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EDUCATION

Nanjing University of Information Science and Technology

Sept 2023 – Jul 2027 (Expected)

B.S. in Atmospheric Science (Hydrometeorology)

Nanjing, China

NUIST ranked #1 in China and #8 globally for Atmospheric Sciences

- Weighted average score: 85.53/100 | GPA: 3.57/4.0 | Rank: 6/51 (top 12%)
- Research affiliate of the **Key Laboratory of Hydrometeorological Disaster Mechanism and Warning of Ministry of Water Resources** since freshman year.
- Relevant coursework: Probability & Statistics (92), Remote Sensing for Hydrometeorology (94), Hydrometeorology Modeling (93), Numerical Weather Prediction (92), Python Programming (90).

AREA OF INTEREST

Remote Sensing · Land–Atmosphere Coupling · Extreme Events · WRF Modeling

PUBLICATIONS

- [1] Deng, K., Huang, Z.*, (2026). **Area expansion dominates rising global three-dimensional terrestrial water storage drought severity** *Geophysical Research Letters*. (to be submitted as Letter)
- [2] Deng, K., Huang, Z.*, Li, H., Wu, J., (2026). **Global unequal exposure to intensifying hydrological droughts**. *Journal of Hydrology*. (Major Revision, R2)
- [3] Wang, J.* and Deng, K. (2024). **A Model that Explains the Contrasting SST Trends in the Southern Pacific Ocean**. *STECEP*. doi.org/10.61173/xnag2k18

RESEARCH EXPERIENCE

Global 3D Hydrological Drought Tracking & Mechanism Attribution

Sept 2025 - Present

Independent research | Supervised by Prof. Zhongwei Huang (NUIST)

Nanjing

- Applied EEMD to 23-year GRACE/GRACE-FO TWS, developed a 3D drought clustering algorithm, identifying 672 discrete global drought events with topological splitting/merging.
- Constructed $C_E = E'/(E' - P')$ evaporative contribution framework: 14.8% of grid cells shifted from supply-deficit to compound droughts; area expansion drives 75% of global severity increase ($p < 0.05$).
- Presented at Tsinghua University DESS 2026 National Forum (70 nationwide, English defense); Principal Investigator of the Provincial Undergraduate Training Program on Innovation and Entrepreneurship (XJDC202610300604).

Global Unequal Exposure to Intensifying Hydrological Droughts

Mar 2025 - Sept 2025

Research assistant | Supervised by Prof. Zhongwei Huang (NUIST)

Nanjing

- Computed DSI from GRACE/GRACE-FO CSR RL06 v03 mascon TWS via EEMD; integrated GPW v4 population grids, decomposing exposure trends: climate 64.5%, population 18.2%, interaction 17.3% (severe events: climate 76.1%).
- Revealed 42.78 million people/yr exposure increase concentrated in developing regions; most high-income regions showed stable/declining trends. Completed R2 revision responding to 6 reviewer comments with 5 supplementary sensitivity analyses. Under Major Revision at *Journal of Hydrology*.

Fluctuation Theorem & TCWV Asymmetry

Sept 2024 - Sept 2025

Independent research | Supervised by Prof. Jun Yin (NUIST). Discussed with Prof. Lamberto Rondoni (Politecnico di Torino)

- Extended nonequilibrium fluctuation theorem (Yin et al., 2024, *J. Climate*) from surface temperature to TCWV using ERA5; quantified PDF asymmetry A_{asym} across $\tau = 1-20$ yr smoothing timescales.
- Validated SST–TCWV Clausius-Clapeyron scaling ($R^2 = 0.95$, 7%/K); migrated tail-rescaling framework to precipitation, demonstrating universality. Produced manuscript draft in Typst and LaTeX.

WRF-based Precipitation Sensitivity over Jing–Jin–Ji

Mar 2025 - Jul 2025

Independent research | Supervised by Dr. Mengya Wang (NUIST)

Nanjing

- Configured WRF/WPS with 3 nested domains on Linux HPC; conducted 16 sensitivity experiments perturbing soil moisture ($\pm 10\%$ to $\pm 50\%$) and urban LULC to simulate rapid regional urbanization.
- Evaluated precipitation responses via spatial correlation, RMSE, and threat scores; identified optimal cumulus and microphysics parameterization schemes minimizing simulated biases.

Precipitation–Runoff Prediction in the Yiluo River Basin

Jun 2024 - Mar 2025

Research assistant. Supervised by Prof. Xing Yuan (LabESM, CAS)

Nanjing

- Applied EOF to monthly precipitation interannual increments, extracting 3 PCs explaining >90% variance; screened key hydroclimatic predictors via correlation analysis and stepwise regression.
- Built BiLSTM + interannual increment hybrid model: TCC improved 16%, RMSE reduced 20% vs. statistical downscaling; CSSPv2 basin runoff forecasts achieved NSE = 65% for monthly streamflow.

Contrasting SST Trends in the Southern Pacific Ocean

Mar 2024 - Sep 2024

Student researcher. Supervised by Prof. George Tselioudis (Columbia University / NASA GISS)

Beijing

- Analyzed 29-year multi-source satellite/reanalysis datasets (Copernicus SST, DUACS SSH, CERES, NCEP, NSIDC SIC) to characterize SST trends across the Southern Pacific (80°W–180, 70°S–30°S).
- Derived Ekman transport (U_E , V_E , W_E) from wind stress to attribute the SST dipole; diagnosed a sea ice radiative positive feedback cycle. Published as co-first author in *STECEP* (2024).

INTERNSHIP

Climate Smart Lab, University of Prince Edward Island

Jul 2026 - Oct 2026

Research Intern. Supervised by Prof. Xander Wang (UPEI)

PE, Canada

- Fully funded by Mitacs Globalink and Chinese Government Scholarship (CSC) (≤ 260 awardees nationwide). Conducting land–atmosphere coupling diagnostics and GRACE-based groundwater trend analysis for Prince Edward Island.
- Comparing satellite TWS estimates with local well measurements to assess freshwater sustainability in a groundwater-dependent province.

Leizhou Meteorological Bureau, Guangdong

Jul 2025 - Aug 2025

Intern, Forecasting Division, Guangdong Provincial Meteorological Observation

Zhanjiang

- Participated in frontline monitoring/warning for Typhoon Danas (2025) and Typhoon Wipha (2025), tracking intensification and landfall trajectories; conducted multi-model NWP comparison (ECMWF, CMA, NCEP, TRAMS).
- Calibrated field hydrologic equipment (groundwater wells, flux towers, snow sensors) utilizing telemetry systems, ensuring continuous environmental data collection for disaster forecasting.

SELECTED AWARDS & HONORS

Chinese Government Scholarship: Chinese Scholarship Council (≤ 260 awardees nationwide)

Jan 2026

Mitacs Globalink Scholarship: Mitacs, Canadian national research organization

Jan 2026

Second-Class Academic Scholarship: NUIST (top 20%)

Nov 2024 & Nov 2025

Merit Student: NUIST (top 15%)

Dec 2024

1st Place, Swimming Medley Relay: NUIST (1/23, department level)

May 2025

ADDITIONAL INFORMATION

Programming: Python (5 yrs+), Shell Script (5 yrs+), Fortran (2 yrs), Typst/LaTeX (3 yrs)

Certificates: MIT Deep Learning Lecture Certificate

Tools: Google Earth Engine, Linux (HPC), Git, WRF, MATLAB

Languages: English (Academic Proficiency), Chinese (Mandarin, Cantonese), French (A2)

Activities:

- Oral presentation in English, National Forum for Outstanding Undergraduates. DESS, Tsinghua University Jun 2026
- Maintains a public academic website (othodeng.github.io) sharing study notes from 10+ courses.
- 10-year swimming training.